Serie 6

Vector spaces and span

1. Polynomials. Consider the polynomials

$$\begin{aligned} p_1(x) &= x^3 + x^2 \\ p_2(x) &= x^2 - 2x - 4 \\ p_3(x) &= 3x + 4 \\ p_4(x) &= 2x + 3 \end{aligned}$$

1. Express the polynomial $2x^3 + 3x^2 - 1$ as a linear combination of the p_i, $i = 1, 2, 3, 4$.
2. Calculate the linear span $\mathrm{Sp}(p_1, p_2, p_3, p_4)$.

2. Dimension 2. Let $v \in \mathbb{R}^2 \smallsetminus \{(0,0)\}$ and let $w \in \mathbb{R}^2 \smallsetminus \{(0,0)\}$ be such that $w \neq \alpha v$ for all $\alpha \in \mathbb{R}$.

(a) Show that $\mathrm{Sp}(v, w) = \mathbb{R}^2$.

(b) Show that the only subspaces of $\mathbb{R}^2$ are $\{(0,0)\}$, $\mathrm{Sp}(v)$ for all $v \in \mathbb{R}^2$, and $\mathbb{R}^2$.

3. Subspaces. Do the operations $+$ and $\cap$ on linear subspaces satisfy distributivity? In other words, do the following equations hold for all linear subspaces?

$$\begin{aligned} U \cap (V_1 + V_2) &= (U \cap V_1) + (U \cap V_2) \\ U + (V_1 \cap V_2) &= (U + V_1) \cap (U + V_2) \end{aligned}$$

If not, is at least one inclusion satisfied?
Recall that for two subspaces V_1 and V_2 of a vector space V,

$$V_1 + V_2 = \{u + v \mid u \in V_1,\ v \in V_2\}.$$

4. Subspaces and equations. Let K be a field. Fix $x \in K^n$ and $b \in K^m$. Define

$$U := \{A \in M_{m \times n}(K) \mid A \cdot x = b\}.$$

For which values of x and b is $U \subseteq M_{m \times n}(K)$ a linear subspace?

5. Polynomials. Prove that $K[x]$ is *not* finite-dimensional over K.

6. Sequences.

(a) Let K_0^∞ be the set of finitely-supported sequences, i.e.

$$K_0^\infty = \{(a_0, a_1, a_2, \dots, a_n, \dots) \mid \forall i \in \mathbb{N} : \ a_i \in K \wedge \exists N \geqslant 0, \forall n \geqslant N : a_n = 0\}.$$

Write the smallest (with respect to inclusion) generating subset $E \subsetneq K_0^\infty$ that you can think of and justify your answer.

(b) Do the same for

$$K_{cst}^\infty := \{(a_0, a_1, a_2, \dots, a_n, \dots) \mid \forall i \in \mathbb{N} : a_i \in K \wedge \exists c \in K, \exists N \geqslant 0, \forall n \geqslant N : \ a_n = c\},$$

the set of eventually constant sequences.